

Physical Security Control Plan

Company Scenario

This Physical Security Control Plan continues the cybersecurity improvement program developed for the same logistics company used in the previous Security Audit and Risk Assessment projects.

Following the findings and recommendations from those assessments, the company is now preparing to bid for a government contract that requires all confidential information to remain onsite. The contract strictly prohibits employees from transporting confidential information using USB flash drives or other removable storage devices.

To satisfy these contractual requirements, the company must strengthen its physical security controls to protect employees, visitors, workstations, networking equipment, servers, and sensitive business information from unauthorized access, theft, tampering, and physical damage.

The first floor includes the reception area, shared office workspace, IT office, meeting room, records storage room, server room, and emergency exit.

The objective of this Physical Security Control Plan is to recommend practical physical and environmental security controls that improve access management, reduce security risks, and protect the organization's critical information assets while supporting compliance with government security requirements.

Step 1: Physical Security Control Plan

The following physical security controls will be implemented on the first floor to strengthen the organization's physical security posture and ensure compliance with government contract requirements.

Physical Security Controls

1. Electronic Access Control System

- Installed at the entrance to the IT Office and Server Room.
- Access will be granted only to authorized personnel using access cards or biometric authentication.

2. Closed-Circuit Television (CCTV) Surveillance

- Cameras will be installed at the main entrance, reception area, shared workspace, hallway leading to the server room, and emergency exit.
- All recordings will be securely stored and monitored by authorized security personnel.

3. Security Guard and Visitor Management Checkpoint

- A security checkpoint will be established at the reception area.
- All visitors must present valid identification, sign the visitor log, and receive a temporary visitor badge before entering the building.

Physical Security Layout Plan

The first-floor physical security layout is designed to strengthen access control, monitor sensitive areas, and reduce the risk of unauthorized access to confidential information.

The proposed layout includes:

A security guard checkpoint and visitor management desk at the main entrance.

CCTV surveillance cameras covering the reception area, shared workspace, hallway to the server room, and emergency exit.

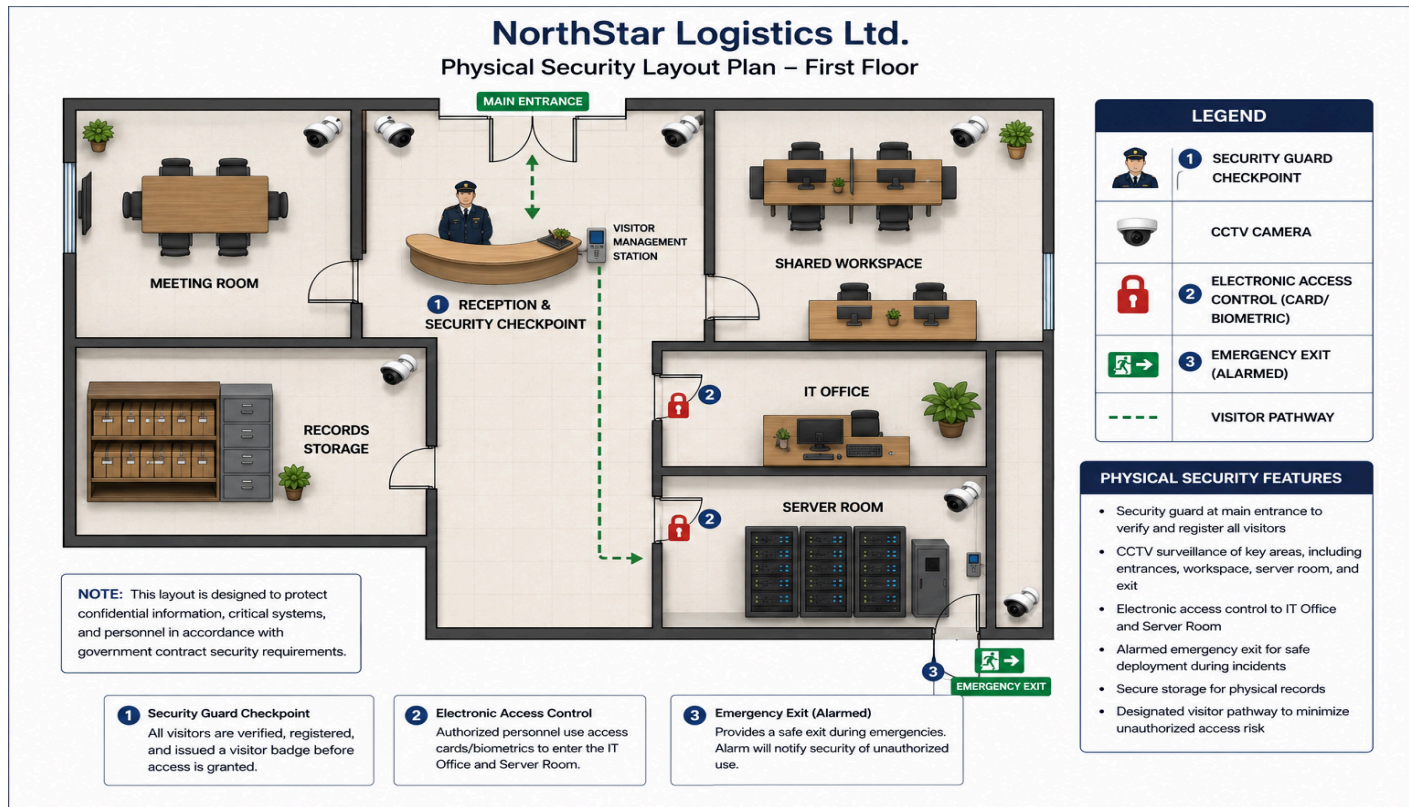
Electronic access control (card or biometric access) protecting the IT Office and Server Room.

Clearly marked emergency exits that remain alarmed and monitored.

Secure records storage with restricted access.

Clearly designated visitor pathways to prevent unauthorized movement into restricted work areas.

A floor plan illustrating the placement of these physical security controls will be inserted below.



Step 2

1. Electronic Access Control System

Location: IT Office and Server Room

Reasoning:

The IT Office and Server Room contain critical business systems, networking equipment, and confidential organizational data. Installing an electronic access control system ensures that only authorized personnel can enter these restricted areas. Using access cards or biometric authentication also creates an audit trail, allowing the organization to monitor who accessed these areas and when.

2. CCTV Surveillance Cameras

Location: Reception Area, Shared Workspace, Hallway to the Server Room, and Emergency Exit

Reasoning:

Strategically placing CCTV cameras provides continuous monitoring of sensitive locations and acts as a deterrent against theft, vandalism, and unauthorized access. The cameras also provide valuable evidence during security investigations while improving overall situational awareness for security personnel.

3. Security Guard and Visitor Management Checkpoint

Location: Main Entrance and Reception Area

Reasoning:

Positioning a security guard at the building entrance ensures that every visitor is properly identified, registered, and issued a visitor badge before entering the facility. This reduces the risk of unauthorized individuals accessing sensitive areas while strengthening compliance with the government's onsite data protection requirements.

Step 3**1. Fire Detection and Clean Agent Fire Suppression System**

Location: Server Room and IT Office

Reasoning:

A clean agent fire suppression system quickly detects and suppresses fires without damaging sensitive electronic equipment. Unlike traditional water sprinklers, clean agents protect servers, networking devices, and storage systems while minimizing downtime and data loss.

2. Uninterruptible Power Supply (UPS) and Backup Generator

Location: Server Room

Reasoning:

A UPS provides immediate backup power during electrical outages, allowing critical systems to continue operating until the backup generator starts. This prevents unexpected system shutdowns, reduces the risk of data corruption, and supports business continuity during power failures.

3. Temperature and Humidity Monitoring System

Location: Server Room

Reasoning:

Maintaining appropriate temperature and humidity levels helps prevent overheating, equipment failure, and moisture-related damage to critical IT infrastructure. Continuous environmental monitoring allows the organization to detect abnormal conditions early and take corrective action before systems are affected.

Conclusion

The recommended physical and environmental security controls strengthen the organization's ability to protect confidential information, critical systems, and personnel from unauthorized access and physical threats. Implementing these controls supports compliance with government contract requirements while improving the company's overall physical security posture, operational resilience, and business continuity.

